Taylor expansion of the optimal winning probability $w_{n,p}$ about $p = \frac{1}{2}$

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1 Setting

We consider the all-heads-wins game: start with n coins, each landing heads independently with probability $p \in (0,1)$; in every round all remaining coins are tossed and at least one must be set aside (never tossed again); the game is won iff every set-aside coin shows heads. Let $w_{n,p}$ be the winning probability under optimal play, with $w_{0,p} := 1$. The Bellman recursion is

$$w_{n,p} \;=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j \, M_{n,j}(p), \qquad M_{n,j}(p) \;:=\; \max_{j \le m \le n-1} w_{m,p}. \tag{1}$$

We write w_n for $w_{n,p}$ when p is understood.

At $p = \frac{1}{2}$ every strategy gives value $\frac{1}{2}$, hence $w_{n,\,1/2} = \frac{1}{2}$ for all $n \ge 1$.

Caveat on regularity. The function $p \mapsto w_{n,p}$ is continuous but typically *not* differentiable at $p = \frac{1}{2}$ once $n \ge 3$: the max in (1) is a two-sided maximum of polynomial pieces, so there is a corner at $p = \frac{1}{2}$ whose left- and right-derivatives disagree in general. We therefore introduce *one-sided* Taylor expansions

$$w_{n,\,1/2+h} \;=\; \tfrac{1}{2} + \alpha_n^{\bullet} h + \beta_n^{\bullet} h^2 + \gamma_n^{\bullet} h^3 + \delta_n^{\bullet} h^4 + O(h^5), \qquad \bullet \in \{\text{left}, \text{right}\}, \tag{2}$$

valid for $h \uparrow 0$ ($\bullet = \text{left}$) and $h \downarrow 0$ ($\bullet = \text{right}$) respectively.

2 Polynomials for small n

At $p = \frac{1}{2}$ every strategy is optimal, so the fully optimal value agrees with strategy B's value on a neighbourhood. Iterating $b_n = p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j b_j$ with $b_0 = 1$ gives

$$\begin{aligned}
w_1(p) &= p, \\
w_2(p) &= 3p^2 - 2p^3, \\
w_3(p) &= 13p^3 - 27p^4 + 21p^5 - 6p^6, \\
w_4(p) &= 75p^4 - 316p^5 + 606p^6 - 664p^7 + 432p^8 - 156p^9 + 24p^{10}.
\end{aligned}$$

The degree of w_n grows like $\binom{n+1}{2}$; these representations match the optimal w_n on a two-sided neighbourhood of $p = \frac{1}{2}$ only for $n \le 2$, and on a left neighbourhood for those n where strategy B remains optimal.

3 Right-hand expansion ($p > \frac{1}{2}$)

By Manuscript Lemma 0.3, strategy A is optimal for $p > \frac{1}{2}$, so $w_{n,p} = a_{n,p}$ where

$$a_n \;=\; p^n + \big(1 - p^n - (1-p)^n\big) a_{n-1}, \qquad a_0 = 1.$$

Differentiating at $p = \frac{1}{2}$ and using $p^n|_{1/2} = (1-p)^n|_{1/2} = 1/2^n$ yields

$$\alpha_n^{\text{right}} \;=\; \frac{n}{2^{n-1}} + \Big(1 - \frac{1}{2^{n-1}}\Big) \alpha_{n-1}^{\text{right}}, \qquad \alpha_1^{\text{right}} = 1. \tag{3}$$

Proposition 1. *$(\alpha_n^{right})_{n\geq 1}$ is strictly increasing and converges to a finite limit α_∞^{right}.*

Proof. The forced positivity of $n/2^{n-1}$ and the contraction factor $1-1/2^{n-1} \in [0,1)$ force $\alpha_n^{\text{right}} - \alpha_{n-1}^{\text{right}} = n/2^{n-1} - \alpha_{n-1}^{\text{right}}/2^{n-1}$, which one checks is strictly positive by induction using $\alpha_n^{\text{right}} < n$ (immediate from (3)). Boundedness ($\alpha_n^{\text{right}} \leq n$ for all n is loose; a sharper bound follows from Lemma 0.3 together with $w_{n,p} \leq 1$) gives convergence. □

Numerically,

$$\alpha_1^{\text{right}} = 1, \quad \alpha_2^{\text{right}} = \tfrac{3}{2}, \quad \alpha_3^{\text{right}} = \tfrac{15}{8}, \quad \alpha_4^{\text{right}} = \tfrac{137}{64}, \quad \alpha_5^{\text{right}} = \tfrac{2375}{1024}, \quad \cdots \quad \alpha_\infty^{\text{right}} \approx 2.607.$$

4 Left-hand expansion ($p < \frac{1}{2}$)

For $h \uparrow 0$ the quantity $M_{n,j}$ in (1) satisfies

$$M_{n,j}\big(\tfrac{1}{2}+h\big) \;=\; \tfrac{1}{2} \;+\; \min_{j\leq m\leq n-1} \alpha_m^{\text{left}} \cdot h \;+\; O(h^2),$$

because among $w_m(\frac{1}{2}+h) = \frac{1}{2} + \alpha_m^{\text{left}} h + \cdots$ (for $h<0$), the maximum is attained where α_m^{left} is *smallest*. Substituting into (1) and using the combinatorial identity

$$\sum_{j=1}^{n-1} \binom{n}{j} (n-2j) \;=\; 0$$

(obtained from $\sum_j \binom{n}{j} = 2^n$ and $\sum_j j\binom{n}{j} = n\,2^{n-1}$) yields the recursion

$$\boxed{\alpha_n^{\text{left}} \;=\; \frac{n}{2^{n-1}} + \frac{1}{2^n}\sum_{j=1}^{n-1}\binom{n}{j} \min_{j\leq m\leq n-1} \alpha_m^{\text{left}}, \qquad \alpha_1^{\text{left}} = 1.} \tag{4}$$

Numerical values (to 18 digits, via (4) with mpmath):

n	α_n^{left}
1	1.000000000000000000
2	1.500000000000000000
3	1.687500000000000000
4	1.734375000000000000
5	1.735839843750000000
6	1.729385375976562500
7	1.722763895988464355
8	1.717139652930200100
9	1.712851968293762180
10	1.709751579242984930
$\vdots$	$\vdots$
∞	≈ 1.703

Proposition 2. *The sequence $(\alpha_n^{left})_{n\geq 1}$ is strictly increasing on $\{1,\dots,5\}$, strictly decreasing on $\{5,6,\dots\}$, and converges to a limit $\alpha_\infty^{left} \approx 1.703$. In particular it has a global maximum at $n=5$.*

Numerical verification of Proposition 2 up to $n = 20$ is straightforward and agrees with the table above.

Corollary 3 (first-order shape of $n \mapsto w_{n,p}$ for $p \lesssim \frac{1}{2}$)**.** *For every sufficiently small $\varepsilon > 0$ there exists $h_0 < 0$ such that for all $h \in (h_0, 0)$,*

$$w_1 \;>\; w_2 \;>\; w_3 \;>\; w_4 \;>\; w_5 \;<\; w_6 \;<\; w_7 \;<\cdots,$$

up to corrections of order $|h|^2$. In particular $n \mapsto w_{n,p}$ is valley-shaped at first order with a unique strict local minimum at $n=5$ and no *strict local maximum at first order.*

Proof. $w_{n,\,1/2+h} - w_{n-1,\,1/2+h} = (\alpha_n^{\text{left}} - \alpha_{n-1}^{\text{left}})\,h + O(h^2)$. The sign of the difference is opposite to the sign of $\Delta\alpha_n := \alpha_n^{\text{left}} - \alpha_{n-1}^{\text{left}}$. By Proposition 2, $\Delta\alpha_n > 0$ for $n = 2, \ldots, 5$ and $\Delta\alpha_n < 0$ for $n \geq 6$, which gives the claimed chain of inequalities for $h < 0$ small. □

Remark. Corollary 3 says that local maxima of $n \mapsto w_{n,p}$ for $p < \frac{1}{2}$ are *not* visible at first order in $h = p - \frac{1}{2}$. They arise only once $|h|$ is large enough that higher-order Taylor terms contribute; this matches the numerical observation that the first local maximum of $(w_{n,p})$ appears at increasingly large n as $p \to \frac{1}{2}^-$ (e.g. at $n = 154$ for $p = 0.49$, but at $n = 9$ for $p = 0.42$).

5 Higher-order coefficients (strategy B)

For n up to the threshold where $w_{n,p} = b_{n,p}$ on a left neighbourhood of $p = \frac{1}{2}$, the Taylor expansion (2) has coefficients equal to those of the polynomial $b_n(p)$, obtained by straight differentiation. Compactly, the first four coefficients for $n = 1, \ldots, 8$ are:

n	α_n^{B}	β_n^{B}	γ_n^{B}	δ_n^{B}
1	1	0	0	0
2	3/2	0	−2	0
3	27/16	−3/8	−7/2	3
4	111/64	−25/32	−15/4	7
5	3555/2048	−535/512	−1735/512	295/32
6	113337/65536	−19167/16384	−3045/1024	39165/4096
7	14471919/8388608	−5067741/4194304	−5624815/2097152	9399929/1048576
8	462775983/268435456	−161718519/134217728	−168136367/67108864	274696555/33554432

so that, for those n,

$$w_{n,\,1/2+h} \;=\; \tfrac{1}{2} + \alpha_n^{\text{B}} h + \beta_n^{\text{B}} h^2 + \gamma_n^{\text{B}} h^3 + \delta_n^{\text{B}} h^4 + O(h^5).$$

Remark. The first-order coefficients α_n^{B} coincide with α_n^{left} as long as the minimum in (4) is attained at the *left* endpoint ($m = j$) for every j, which is the case precisely when strategy B is optimal at step n. Once that fails (i.e. once the sequence α_m^{left} stops being monotone increasing), α^{B} and α^{left} diverge. The first such discrepancy occurs between $n = 5$ and $n = 6$, which is exactly where the numerical monotonicity of the optimal w-sequence first breaks for p close to $\frac{1}{2}$.

6 Summary

- $w_{n,p}$ is C^0 but typically only piecewise C^∞ near $p = \frac{1}{2}$; left/right derivatives differ for $n \geq 3$.
- Right-side: α_n^{right} strictly increases, consistent with Lemma 0.3.
- Left-side: α_n^{left} is valley-shaped (increasing then decreasing), so at first order in $h = p - \frac{1}{2} < 0$ the sequence $(w_{n,p})_n$ has a strict local minimum at $n = 5$ and no local maximum.
- Local maxima of $(w_{n,p})$ for $p < \frac{1}{2}$ are therefore a *non-perturbative* phenomenon near $p = \frac{1}{2}$: they appear only at n and p at which higher-order Taylor terms overpower the first-order valley structure.